

Question Number	Answer	Additional Guidance	Mark
2a	- suitable risk identified (1) - reduction of risk identified to the volunteer (1)	Mark first risk identified e.g. risk of infection / carbon dioxide not absorbed/ CO₂ poisoning / breathing difficulties / named condition / {soda lime / carbon dioxide absorber} is irritant e.g. sterilise or replace mouthpiece (after each use) / replace carbon dioxide absorber / medical questionnaire / do not take part if at risk / wear gloves / goggles Ignore clean unqualified	2

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2b	- answer in the correct range (1) - use of appropriate units (1)	Range of 6.1 to 6.5 $\times 10^{-3}$ (if the units are $\text{dm}^3 \text{s}^{-1}$) Allow any number of decimal places Allow minus values <table><tr><th>Units</th><th>Acceptable range</th></tr><tr><td>$\text{dm}^3 \text{s}^{-1}$</td><td>$6.1 \times 10^{-3} - 6.5 \times 10^{-3}$</td></tr><tr><td>$\text{dm}^3 \text{min}^{-1}$</td><td>$3.7 \times 10^{-1} - 3.9 \times 10^{-1}$</td></tr><tr><td>$\text{cm}^3 \text{s}^{-1}$</td><td>6.1 – 6.5</td></tr><tr><td>$\text{cm}^3 \text{min}^{-1}$</td><td>366 - 390</td></tr></table> Not dm^3/s or dm^3/min Allow per second or per minute Allow $\text{m}^3 \text{s}^{-1}$ acceptable range is 6.1 to 6.5×10^{-6} $\text{m}^3 \text{min}^{-1}$ acceptable range is 3.7 to 3.9×10^{-4}	Units	Acceptable range	$\text{dm}^3 \text{s}^{-1}$	$6.1 \times 10^{-3} - 6.5 \times 10^{-3}$	$\text{dm}^3 \text{min}^{-1}$	$3.7 \times 10^{-1} - 3.9 \times 10^{-1}$	$\text{cm}^3 \text{s}^{-1}$	6.1 – 6.5	$\text{cm}^3 \text{min}^{-1}$	366 - 390	
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2ci	An answer that includes two of the following points biotic • age (1) • {BM1 / mass / height} (1) • medical history (1) • sex (1) • {lung capacity / named example of lifestyle factor affecting this} (1)	List rule applies Allow weight for mass obesity eg. asthma, cystic fibrosis, heart disease Allow gender eg smoking, exercise, playing wind instrument etc	2

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2cii	• description of an appropriate method of control (1)	ECF for inappropriate variable chosen	1

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2d	An answer that includes the following points • chemoreceptors detect (change of) {pH / carbon dioxide concentration/level} (in blood) (1) • (nerve) impulses sent to {intercostal muscles / diaphragm} (1) • rate of breathing increases with {increased CO₂ / lower pH} (1)	Allow converse statement	3

